Two-channel signatures of the E_7 spectrum in a blockaded Rydberg chain

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The thermal deformation of the tricritical Ising conformal field theory is an integrable theory with seven E_7 particles. In a blockaded Rydberg chain, lattice momentum separates its odd and even particles: a staggered detuning probe couples to σ and a uniform one to ε. Exact diagonalization of the strictly blockaded chain with a $1/r^6$ tail, up to $L = 36$, locates a direction-dependent parameter region in which the four light particles A_1–A_4 appear as two sharp peaks per channel, with gap ratios within 3% of E_7. This agreement changes by less than one percentage point across the reported 1.7% calibration uncertainty of the interaction. On open chains of 19–27 atoms, the four resonances are consistent with standing waves of the bulk particles, with the largest deviation for A_4; a tapered drive suppresses A_1 and A_2 cross-channel leakage below 0.03%, but leaves 7.5% and 1.9% for A_3 and A_4. The shallow bound state A_4 carries the slowest finite-size corrections and sets the resolution requirement. The heavy particles A_5–A_7 are not resolved.

Introduction.—Relevant perturbations of a two-dimensional conformal field theory (CFT) can produce integrable massive theories with spectra fixed by exceptional Lie algebras. The magnetic perturbation of critical Ising theory realizes E_8 [1], with experimental signatures in $CoNb_2O_6$ [2] and Rydberg arrays [3]. The leading thermal perturbation of the tricritical Ising (TCI) CFT, with central charge $c = 7/10$, instead realizes E_7 [4–6]. Writing $r_a = m_a/m_1$, the seven ratios are

$$(r_1, \ldots, r_7) = (1,\ 1.2856,\ 1.8794,\ 1.9696,\ 2.5321,\ 2.8794,\ 3.7017). \quad (1)$$

with $\mathbb{Z}_2$ parities $(-,+,-,+,+,-,+)$ in the disordered phase [7]. An experimental identification of this spectrum remains an open problem [8].

Rydberg chains host Ising criticality, and an effective attraction produces a tricritical point [9, 10]. Sun *et al.* [11] reached it by preparing the upper edge of the blockaded sector, equivalent in the projected model to an attractive ground-state problem, and measured finite-size CFT spectra. This motivates spectroscopy of the neighboring gapped regime.

The field-theory inputs are established: the Rydberg density–CFT operator correspondence [10], the E_7 form factors and selection rules [6, 12], and their threshold structure. Király and Lencsés [13] have also obtained lattice-quench spectra compatible with the even particles A_2, A_4, A_5 in the Blume–Capel chain, including form-factor amplitude comparisons. That study uses A_2 to calibrate the mass scale; it does not independently resolve A_1. Here we show that a Rydberg chain gives access to both parity sectors, so that odd and even light particles can be compared on one platform. We map the parameter region where four light peaks match E_7, identify the standing-wave structure of open-chain resonances, and quantify the cross-channel leakage and the finite-size constraint set by A_4.

Model and field content.—We study the projected Hamiltonian

$$H = \sum_j \left[\frac{\Omega}{2} P_{j-1} X_j P_{j+1} - \Delta n_j\right] + \sum_{i<j,\, d_{ij}\geq 2} V_2 (2/d_{ij})^6 n_i n_j. \quad (2)$$

with the constraint $n_j n_{j+1} = 0$, $P_j = 1 - n_j$, and $V_2 < 0$. The distance d_{ij} is $|i-j|$ for open chains and the shortest distance on a periodic chain; each pair is counted once. We set $\hbar = \Omega = a = 1$, where a is the lattice spacing. All allowed interaction pairs are retained within each finite chain.

Equation (2) is the strict-blockade approximation to the sign-reversed experimental problem, not the full experimental Hamiltonian. It omits the finite-blockade correction H_2 of Ref. [11], which contains diagonal shifts and correlated hopping of order Ω^2/V_1. We compute ground-state response on open chains without boundary detuning.

In the long-wavelength theory, one-site translation exchanges the two density-wave patterns and implements the internal $\mathbb{Z}_2$ together with spatial translation [10]. The density has the expansion

$$n_j \simeq \bar{n} + c_\varepsilon \varepsilon + c_{\varepsilon'} \varepsilon' + (-1)^j (c_\sigma \sigma + c_{\sigma'} \sigma') + \cdots . \quad (3)$$

Thus zero-momentum even particles occur near lattice momentum 0, and odd particles near π. This identification uses bulk translation symmetry; open-chain reflection parity is a different quantum number.

Both Δ and V_2 generate the even relevant perturbations $g_\varepsilon \varepsilon + g_{\varepsilon'} \varepsilon'$, with RG eigenvalues 9/5 and 4/5. For a straight path of distance t from TCI with a nonzero linear component in g_ε,

$$\eta = \frac{g_{\varepsilon'}}{|g_\varepsilon|^{4/9}} \propto t^{5/9}, \qquad m_1 \propto t^{5/9}. \quad (4)$$

On the disordered side such paths approach the thermal E_7 theory as $t \to 0$ if the system size is increased simul-

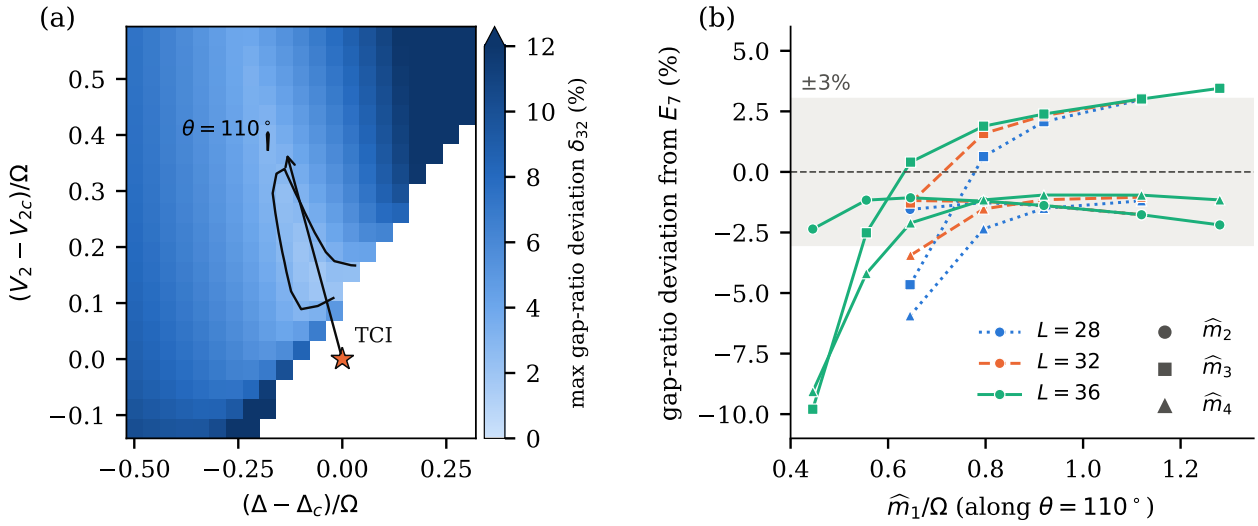


FIG. 1. Finite-chain spectral matching region. (a) $\delta_L = 100 \max_{a=2,3,4} |(\widehat{m}_a/\widehat{m}_1)/r_a - 1|$ at $L = 32$; the contour marks 3%. Grid spacings are 0.04Ω in Δ and 0.035Ω in V_2. Blank cells fail the numerical filters $L\widehat{m}_1/\Omega \geq 18$ and $\widehat{m}_2 \geq \widehat{m}_1/2$; these filters do not establish phases or convergence. The star estimates TCI; the arrow marks $\theta = 110^\circ$. (b) Individual gap-ratio deviations along that direction. Symbols identify the candidate levels and line styles identify sizes. The band is ±3%; lines guide the eye. Neither panel is an experimental error budget.

taneously. The growing correlation length limits this approach at fixed L. We have not calibrated the map from lattice parameters to $(g_\varepsilon, g_{\varepsilon'})$; a region of small mass-ratio errors need not identify the pure thermal trajectory or separate competing corrections.

Probes and numerical method.—For a periodic ground state, the connected density response is

$$S_k(\omega) = \frac{1}{L} \sum_{n \neq 0} \left| \left\langle n \left| \sum_j e^{ikj} n_j \right| 0 \right\rangle \right|^2 \delta[\omega - (\mathcal{E}_n - \mathcal{E}_0)]. \quad (5)$$

Uniform and staggered probes select $k = 0$ and π exactly. In the E_7 limit, Eq. (3) couples them principally to ε and σ, respectively. The proposed $k = \pi$ coupling to σ already appears in Ref. [11]. For a pure continuum operator $\mathcal{O}$, one-particle weights obey $w_a \propto |F_a^{\mathcal{O}}|^2/m_a$. Known form factors give $w_3/w_1 = 0.067$, $w_6/w_1 = 0.0030$ for σ, and $w_4/w_2 = 0.143$, $w_5/w_2 = 0.038$, $w_7/w_2 = 0.0012$ for $\Theta \propto \varepsilon$ [6, 12]. Absolute interchannel amplitudes are not universal.

We use sparse exact diagonalization in the constrained space for periodic $L \leq 36$ and open $L = 19, 23, 27$. Spectra use reorthogonalized Lanczos iterations (up to 400 for the periodic spectra and 250 for the open spectra shown). Small-system checks compare momentum-sector energies and weights with full-space diagonalization to numerical precision. They do not establish convergence of the large-system continuum.

We denote the first two gaps in the $k = \pi$ sector by $\widehat{m}_1(L), \widehat{m}_3(L)$ and the first two excited gaps at $k = 0$ by $\widehat{m}_2(L), \widehat{m}_4(L)$. These are candidate light-particle levels, not independently extrapolated masses. Throughout the parameter scan, the labels follow energy ordering within each sector. For open chains, ω_a° denotes a candidate resonance energy selected by its dominant probe weight.

Simultaneous crossings of L times the lowest odd and lowest excited even gaps locate a finite-size TCI estimate. The $L = 28/32$ crossing gives $(\Delta_c, V_{2c}) = (-1.5011, -1.6260)$, within 10^{-3} of the 24/28 estimate and close to the experimental calibration [11]. Fixing $v \simeq 1.90$ using $x_\sigma = 3/40$ gives other rescaled gaps near 0.875 at $k = \pi$, and 0.198, 1.18 at $k = 0$, consistent with $x_{\sigma'} = 7/8$, $x_\varepsilon = 1/5$, $x_{\varepsilon'} = 6/5$. The value of x_σ is an input to this comparison.

Matching region and finite size.—We scan $(\Delta, V_2) = (\Delta_c, V_{2c}) + t(\cos\theta, \sin\theta)$. Figure 1 identifies a finite region of small errors among the three gap ratios. Along $\theta = 110^\circ$, the sampled $L = 36$ points at $t = 0.1, 0.15, 0.2$ have maximal errors below 3%; at $t = 0.3$ the maximum is 3.01%. Nearby directions 90° and 130° also contain matching points. In contrast, the sampled $\theta = 180^\circ$ points never reach 5% accuracy; deviations near 210° can reach 10–25%. This direction dependence concerns the accessible finite-chain window and does not contradict the generic asymptotic limit in Eq. (4).

At $\theta = 110^\circ, t = 0.2$, corresponding to $(\Delta, V_2) = (-1.5695, -1.4381)$, we find $\widehat{m}_1(36) = 0.9196\Omega$ and

$$\frac{(\widehat{m}_2, \widehat{m}_3, \widehat{m}_4)}{\widehat{m}_1} = (1.2678,\ 1.9243,\ 1.9508). \quad (6)$$

The relative deviations are $(-1.39, +2.39, -0.95)\%$. At $t = 0.1$, however, $\widehat{m}_3/\widehat{m}_1$ changes from 1.792 to 1.887

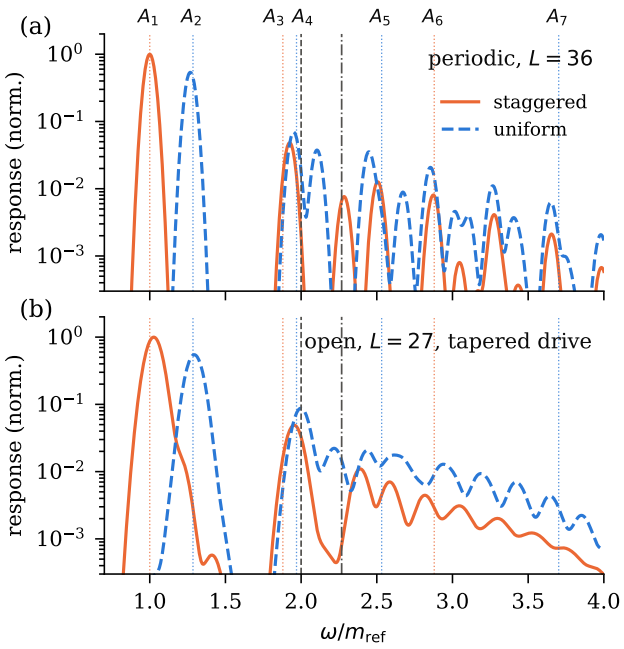


FIG. 2. Response at Eq. (6): (a) periodic $L = 36$, without size averaging; (b) open $L = 27$ with a $\sin^2$ drive envelope. Both use $m_{\rm ref} = \widehat{m}_1(36) = 0.9196\Omega$ and are normalized to their own staggered peak height. Solid orange: staggered; dashed blue: uniform. Only (a) has exact momentum selection. Gaussian standard deviations $0.03m_{\rm ref}$ and $0.05m_{\rm ref}$ illustrate smoothing, not instrumental linewidths. Dotted lines mark ideal E_7 ratios. Gray dashed/dash-dotted lines mark $2m_{\rm ref}$ and $m_{\rm ref} + \widehat{m}_2(36)$: periodic-gap sums, not measured open-chain continuum edges.

between $L = 28$ and 36, illustrating the limitation closer to TCI. At $t = 0.3$ the three ratios change by less than 0.24% over those sizes, but this comparison is not an infinite-volume extrapolation.

The shallow A_4 bound state requires particular care. In the integrable theory, A_4 occurs in the A_1A_1 scattering channel, with

$$\mu_{11}^4 = \sqrt{m_1^2 - m_4^2/4} = m_1 \sin(\pi/18) \simeq 0.174 m_1. \quad (7)$$

Its finite-volume energy can receive a leading bound-state correction proportional to $e^{-\mu_{11}^4 L/v}$, not simply $e^{-m_1 L/v}$ [12, 14]. With the working gap and $v \simeq 1.90$, the E_7 value of μ_{11}^4 gives $v/\mu_{11}^4 \simeq 12$ sites; the binding measured on the chain gives $\simeq 9$ sites. Both exceed $v/m_1 \simeq 2$ sites considerably. In the data, $\widehat{m}_4/\widehat{m}_1$ changes from 1.9400 to 1.9508 between $L = 28$ and 36: only 0.56% of its value, but 22% of the latter binding-energy estimate $2\widehat{m}_1 - \widehat{m}_4$. A mass-ratio tolerance alone therefore does not control near-threshold binding.

At the working point, the periodic weight ratios are $w_4/w_2 = 0.129$ and $w_3/w_1 = 0.0483$. Mixing with σ' contributes an effective relative amplitude proportional to $(c_{\sigma'}/c_\sigma)(m_1 a/v)^{4/5}$ at fixed CFT normalization, since $x_{\sigma'} - x_\sigma = 4/5$. The weights also contain interference, finite-size, and nonintegrable corrections. We do not isolate these sources or treat the pure-field ratios as precise lattice predictions.

Spectra and boundaries.—The periodic response in Fig. 2(a) has two prominent subthreshold peaks in each sector, consistent with the four candidate levels in Eq. (6). Momentum selection is exact. The next visible even-sector level at $L = 36$ is at $2.105m_{\rm ref}$, so its separation from the A_4 candidate is $0.154m_{\rm ref}$, appreciably larger than the gap to the kinematic reference $2m_{\rm ref}$. Discrete scattering levels can thus make a small-system spectrum appear more separated than its infinite-volume counterpart.

For open chains we use $O_{f,q} = \sum_j f_j e^{iqj} n_j$, where $q = 0, \pi$ labels the drive pattern rather than an exact momentum sector. Spectral weights are normalized by $\sum_j f_j^2$. We compare $f_j = 1$ with $f_j = \sin^2[\pi(j+1)/(L+1)]$, $j = 0, \ldots, L-1$. At $L = 27$, the latter gives dominant candidate resonance ratios

$$(\omega_2^{\rm o}, \omega_3^{\rm o}, \omega_4^{\rm o})/\omega_1^{\rm o} = (1.2612,\ 1.9030,\ 1.9488), \quad (8)$$

with $\omega_1^{\rm o} = 0.9438\Omega$, within 2% of E_7. The open-chain energies are shifted upward by the lowest standing-wave momentum. With $E_a(n) = \{\widehat{m}_a^2 + [nv\pi/(L+1)]^2\}^{1/2}$, $v = 1.90$, and the periodic $L = 36$ masses, the $n = 1$ prediction reproduces $\omega_1^{\rm o}$ to 0.02% and $\omega_{2,3}^{\rm o}$ to better than 0.8% at $L = 27$. The largest deviation occurs for A_4: 1.8%, 2.9%, and 5.2% at $L = 27$, 23, and 19. This size dependence is consistent with A_4 having the largest spatial extent, as estimated above.

Table I reports all four channel leakages, defined as the integrated weight at the same candidate eigenenergy in the unintended probe divided by its weight in the intended probe. At $L = 27$, tapering reduces the A_1 leakage from 0.149% to 0.0267%, but the A_3 leakage only from 9.22% to 7.46%, and the A_4 leakage increases from 1.41% to 1.94%. Tapering therefore does not separate all sectors uniformly; at $L = 19$, two nearby uniform resonances also share substantial weight. The strongest satellite, at $1.160\,\omega_1^{\rm o}$ for $L = 27$, is the $n = 3$ standing wave of A_1: under a flat drive, $(L+1)(E^2 - \widehat{m}_1^2)^{1/2}$ equals 16.6, 16.4, and 16.6 at $L = 19$, 23, and 27, about 2.7–2.8 times the $n = 1$ value. The ratio below three is consistent with lattice-dispersion curvature at this momentum. A flat drive gives it 9–13% of the A_1 weight, roughly as expected from its overlap with a uniform profile, and tapering reduces this to 1.7% at $L = 27$. Its origin is thus bulk A_1, not a boundary bound state.

Thresholds and unresolved heavy particles.—In the integrable, translation-invariant theory, the lowest even and odd two-particle thresholds are $2m_1$ (A_1A_1) and $m_1 + m_2$ (A_1A_2). The exact E_7 scattering amplitudes and form factors imply responses proportional to $(\omega - 2m_1)^{1/2}$

TABLE I. Leakage ratios in percent for tapered open-chain probes at the working point. The intended probe is staggered for A_1, A_3 and uniform for A_2, A_4. Labels denote finite-chain candidates.

Candidate	$L = 27$	$L = 19$
A_1	0.0267	0.228
A_2	0.0192	4.53
A_3	7.46	39.4
A_4	1.94	8.98

and $(\omega - m_1 - m_2)^{-1/2}$ in the corresponding channels [12]. Here $S_{11}(0) = -1$ forces the identical-particle form factor to vanish, whereas the relevant A_1A_2 form factor is nonzero at threshold. These are integrable-limit predictions for the matrix-element-weighted response.

At our working point, the next odd-sector spectral level above the two light peaks moves from $2.340\widehat{m}_1$ at $L = 18$ to $2.284\widehat{m}_1$ at $L = 36$, toward the gap sum $1 + \widehat{m}_2/\widehat{m}_1 = 2.268$. The corresponding even-sector sequence moves from $2.296\widehat{m}_1$ at $L = 24$ to $2.105\widehat{m}_1$ at $L = 36$. This is consistent with the two different onsets, but does not determine threshold exponents; on open chains, residual weight remains above A_3.

We do not resolve A_5, A_6, A_7. In the E_7 theory their masses lie above allowed thresholds in their own sectors: A_5, A_7 above $2m_1$, and A_6 above $m_1 + m_2$. Nonintegrable perturbations can open decay channels such as $A_5 \to A_1A_1$ and $A_6 \to A_1A_2$ [7]. Across $L = 18$–36 at $\theta = 110°$, $t = 0.2, 0.3$, the higher spectra do not provide a stable heavy-particle assignment by energy and weight. Finite chains do not measure decay widths, so this neither establishes nor excludes heavy resonances in larger systems.

Experimental requirements.—The protocol prepares a state near Eq. (6) with the demonstrated TCI preparation [11], applies weak tapered uniform and staggered modulations, and compares the two responses. The sign-changing staggered drive needs the two-color (515 and 813 nm) tweezer scheme proposed in Ref. [11], beyond the one-color light shifts used so far.

Spectral resolution is especially consequential for A_4. The finite-chain estimate $2\widehat{m}_1 - \widehat{m}_4 \simeq 0.049 m_{\rm ref}$ is about $2\pi \times 0.25$ MHz for $\Omega = 2\pi \times 5.5$ MHz; in ideal E_7 the relative binding is smaller, $0.0304 m_1$. The illustrative open-chain broadening in Fig. 2 has FWHM $= 2\sqrt{2\ln 2}\,\sigma = 0.118 m_{\rm ref}$, about $2\pi \times 0.60$ MHz. Separating A_4 from the continuum therefore needs a linewidth below this scale; the E_7 threshold suppression of the A_1A_1 response would help in the scaling regime. Because of open-chain mixing, A_3 and A_4 should be assigned by comparing both probes rather than by channel label alone.

The reported 1.7% uncertainty in V_2 [11] amounts to about 0.028Ω. At $\Delta \simeq -1.60\Omega$, the 3% region of Fig. 1(a) extends over at least 0.14Ω in V_2. Near the working point, δ_{32} changes by at most 0.7 percentage points per V_2 grid step and 0.9 per Δ step. Calibration errors of this size therefore move the gap ratios by less than one percentage point in this criterion.

Finite-blockade corrections, preparation, decoherence, and readout are not modeled; they must be included before the four-peak signature is used as an experimental identification.

Conclusion.—Because one-site translation implements the tricritical $\mathbb{Z}_2$, a Rydberg chain can resolve both parity sectors of the E_7 spectrum with uniform and staggered probes. In the strictly blockaded model, we find a direction-dependent region where four light particles appear as two peaks per channel at E_7 ratios within 3%. This agreement is stable against the reported interaction calibration uncertainty. Open-chain resonances follow the bulk masses through standing-wave dispersion; a tapered drive suppresses the dominant satellites, but channel mixing remains particle dependent. The shallow bound state A_4 sets the most demanding finite-size and resolution requirements; the heavy particles remain open. These results specify where to look and which systematics decide an observation.

[Acknowledgments.]

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